

EECS 489 - WN 25

Discussion 7

Intra-AS Routing, Inter-AS Routing, Poisoned Reverse

Announcements

Assignment 3 is out.

- Due **Monday, March 31**
- 3 total late days across A2-A4
- See March 13/14 discussion for tips on using UDP sockets

Intra-AS Routing

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Key Idea: There is one centralized authority within an Autonomous System (AS).

- **Implication:** Not many business considerations when routing; generally the objective is to just minimize the *cost* of traffic.
 - Cost can be many things; often just a proxy for latency/distance.
- **Implication:** Authority can set own policies within AS as they control all the devices
 - Can choose between link-state and distance-vector families of protocols (or even something else)
 - Can choose which specific protocol to use within these protocol families

Intra-AS Routing

Link-State (LS) Routing

- Every router knows local link state; floods local state to all routers in network
 - More, smaller messages
- Implemented using Open Shortest Path First (OSPF) protocol (or IS-IS)
- Based on Dijkstra's Algorithm
- Error is limited, as each node computes own table

Distance-Vector (DV) Routing

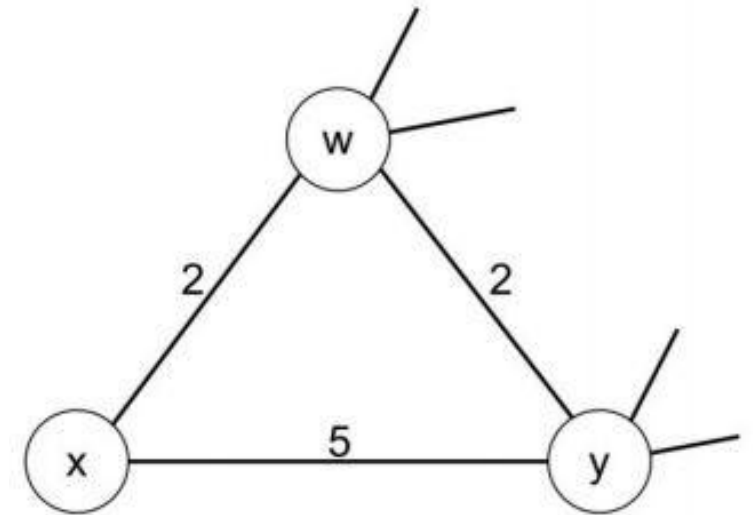
- Every router communicates with its neighbors about global view
 - Fewer, larger messages
- Implemented using Routing Information Protocol (RIP)
- Based on Bellman-Ford Algorithm
- Error can propagate, as nodes rely on the tables of other nodes

Q1 Distance-Vector Situations

Consider this network fragment:

- **w** has least-cost path to **u** of 5
- **y** has least cost path to **u** of 6
- Complete paths from **w** and **y** to **u** not shown
- All links have strictly positive costs

What is **x**'s distance vector for **w**, **y**, and **u**?



$$D_w(u) = 5$$

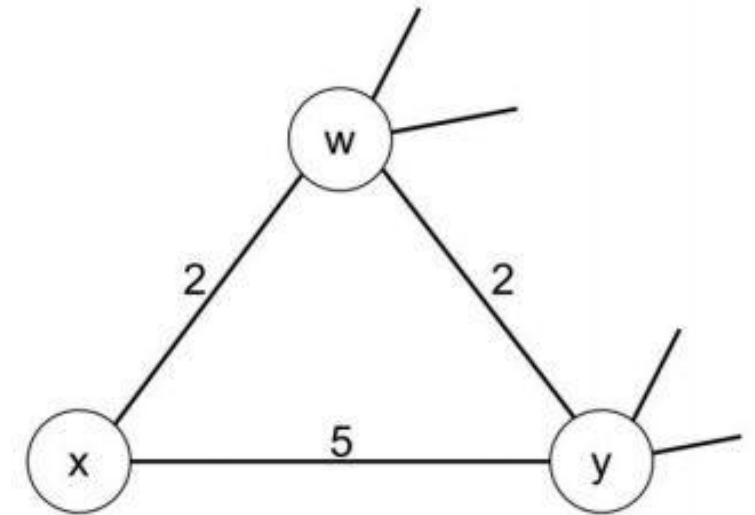
$$D_y(u) = 6$$

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$$D_w(u) = 5$$

$$D_y(u) = 6$$

$$D_x(w) = 2$$

$$D_x(y) = 4$$

$$D_x(u) = 7$$

$$x \rightarrow w \rightarrow y$$

$$x \rightarrow w \rightarrow \dots \rightarrow u$$

Q2: Distance Vector Updates

Suppose that you are a router in a network that uses **distance vector** updates for calculating the shortest path to a destination **with poisoned reverse**. Suppose that your currently calculated distance vector to other routers is

$$\begin{pmatrix} k & u & v & z \\ 5 & 7 & 2 & 10 \end{pmatrix}$$

Router **k** then sends you the following updated distance vector message:

$$\begin{pmatrix} k & u & v & z & m \\ 0 & 3 & 5 & 4 & 7 \end{pmatrix}$$

What is your updated distance vector?

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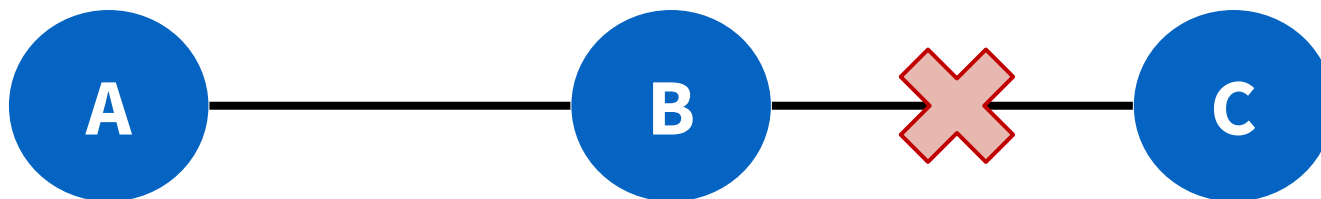
$$\begin{pmatrix} k & u & v & z & m \\ 0 & 3 & 5 & 4 & 7 \end{pmatrix}$$

What is your updated distance vector?

Answer: $\begin{pmatrix} k & u & v & z & m \\ 5 & 7 & 2 & 9 & 12 \end{pmatrix}$

Count-to-Infinity

- Can occur when a link cost increases or fails in Distance-Vector protocols
- A **routing loop** is created when (simplest case):
 - A routes to C through B
 - A advertises to B that it has a route to C, even though this route goes through B
 - The B-C link fails
 - B attempts to route to C through A!



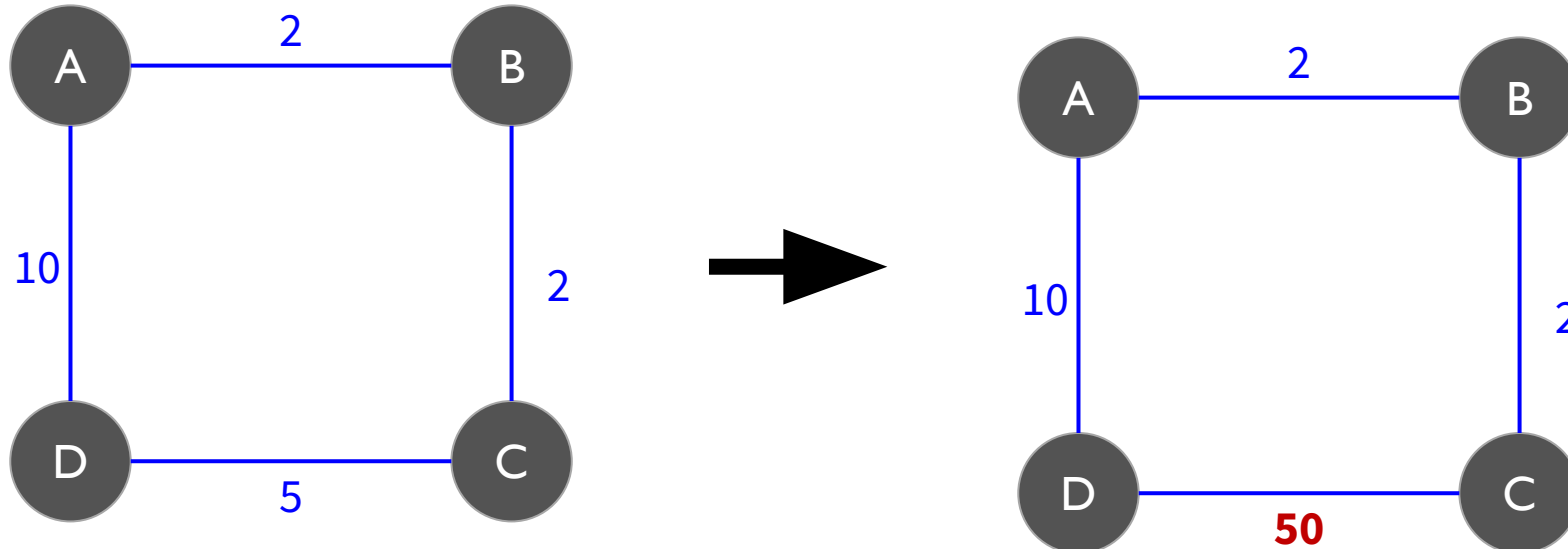
Q3 Distance-Vector Properties

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Yes; the count-to-infinity problem occurs when two nodes decide to route through each other. This can occur when a link becomes disconnected, or its cost increases. **Note:** Not just to infinity; any time we have to iterate to converge.



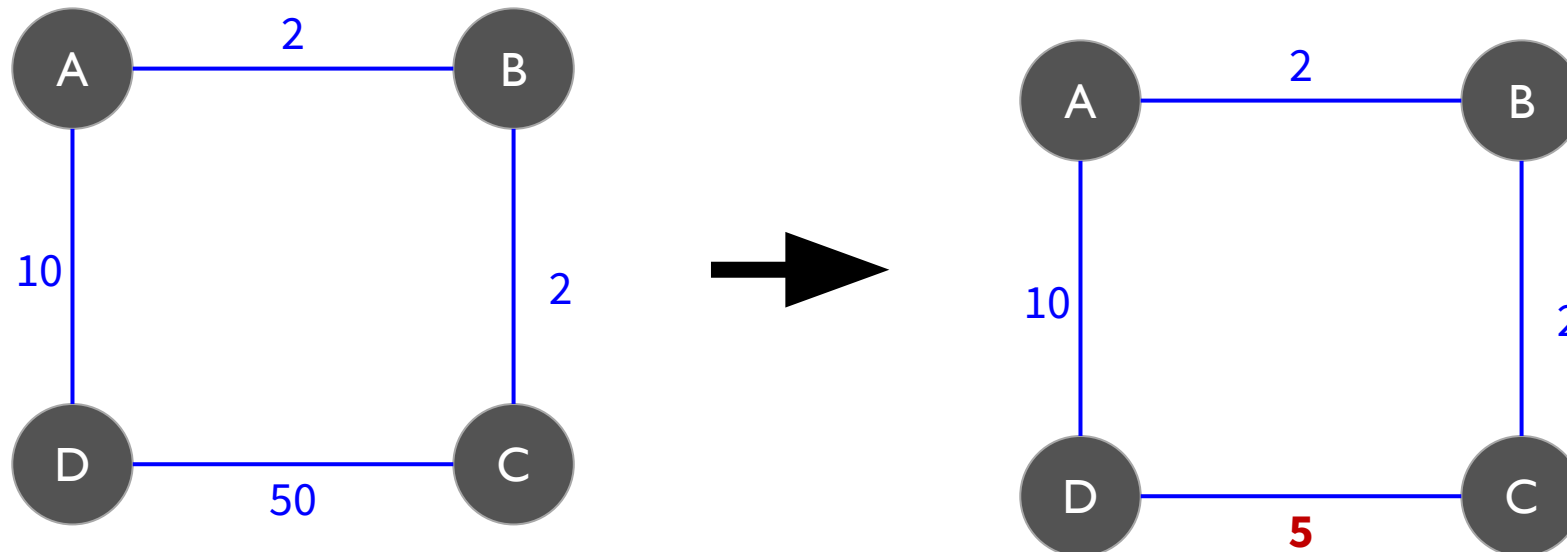
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No. Count-to-infinity is a problem when increasing link cost because traffic is re-routed to nodes that are further from the affected link than the node doing the rerouting. This is not the case with decreasing a link's cost.



Q5 Distance-Vector Properties

Suppose we are using a distance-vector (DV) protocol for routing. Can the count-to-infinity problem ever occur upon **connecting two previously unconnected nodes**?

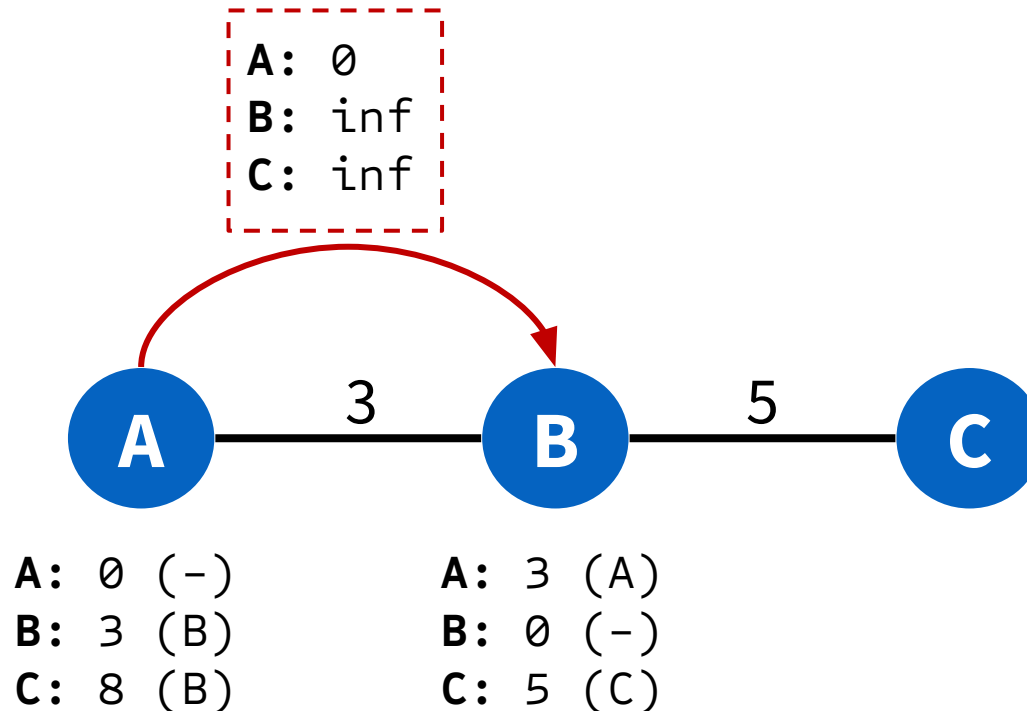
Q5 Distance-Vector Properties

Suppose we are using a distance-vector (DV) protocol for routing. Can the count-to-infinity problem ever occur upon **connecting two previously unconnected nodes**?

Yes; this is morally the same thing as decreasing a link's cost (i.e. decreasing it from $\sim$ infinity to the finite cost of the new link), so the same logic applies as the previous part.

Fixing Count-to-Infinity

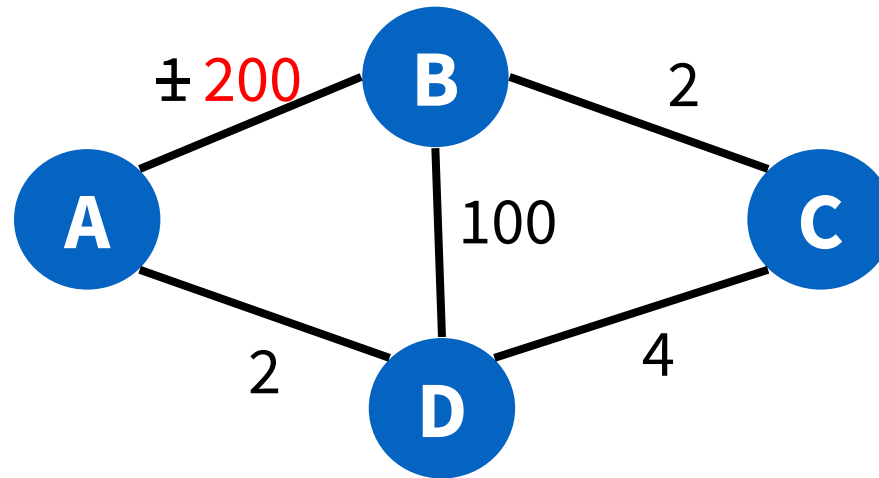
- **Poisoned Reverse:** If A routes to C with first-hop B, A does not advertise A-C path to B (instead, tells B that cost to C is infinity).
 - Prevents routing loops by not allowing interdependence for same destination
 - **Not foolproof – might reduce the problem, but does not fix it!**



Q6 Poisoned Reverse

Consider this network fragment. Suppose the distance-vector routing protocol is being used. Suppose the routing state is stable, when $c(A, B)$ suddenly increases from 1 to 200.

Would count to infinity occur if poisoned reverse **is not** being used?

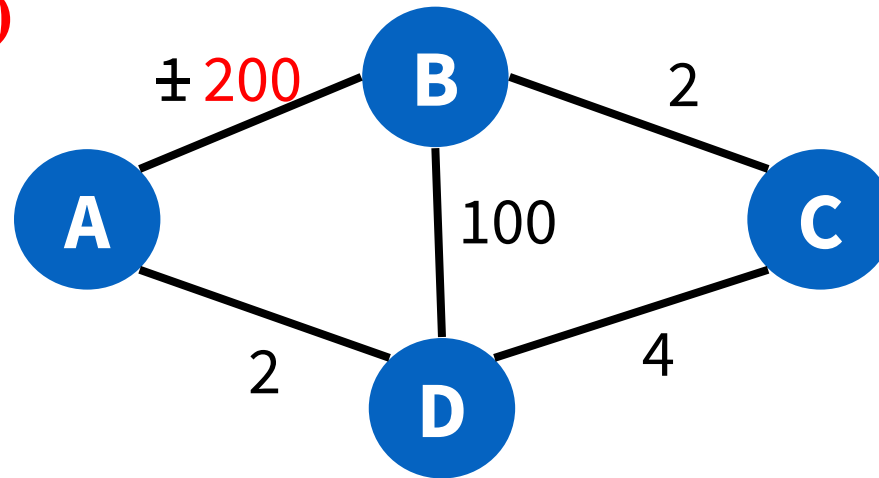


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Yes (A, D routing to B)

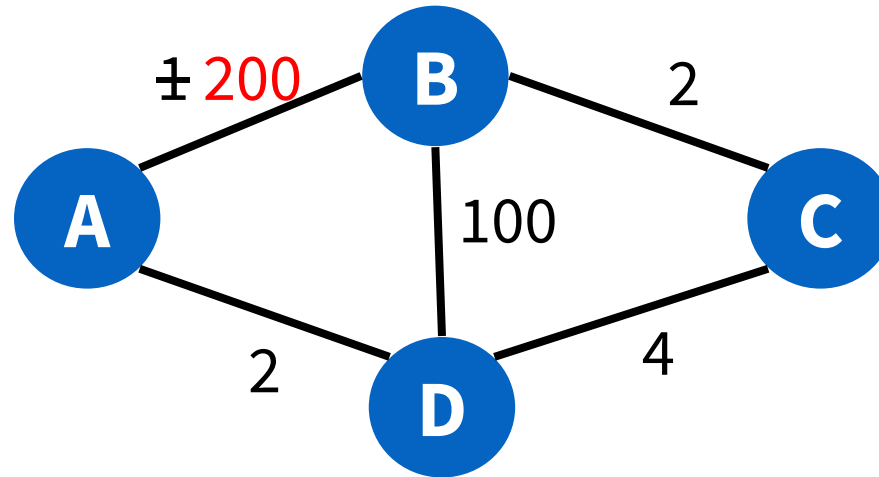


Q7 Poisoned Reverse

Consider this network fragment. Suppose the distance-vector routing protocol is being used. Suppose the routing state is stable, when $c(A, B)$ suddenly increases from 1 to 200.

Would count to infinity occur if poisoned reverse is being used?

No; A will advertise its cost to B as 200, and updates will correctly follow from there.



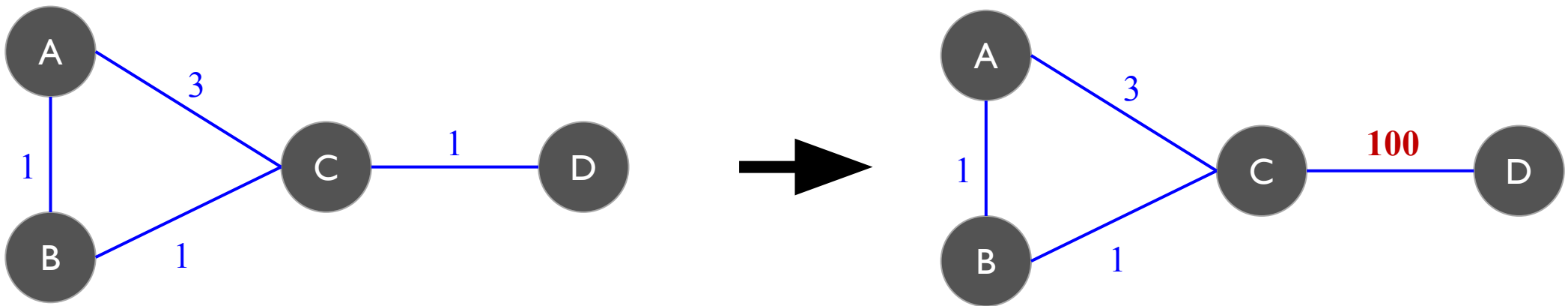
Q8: Poisoned Reverse Failing

Can you come up with a scenario where poisoned reverse fails to prevent a count-to-infinity scenario?

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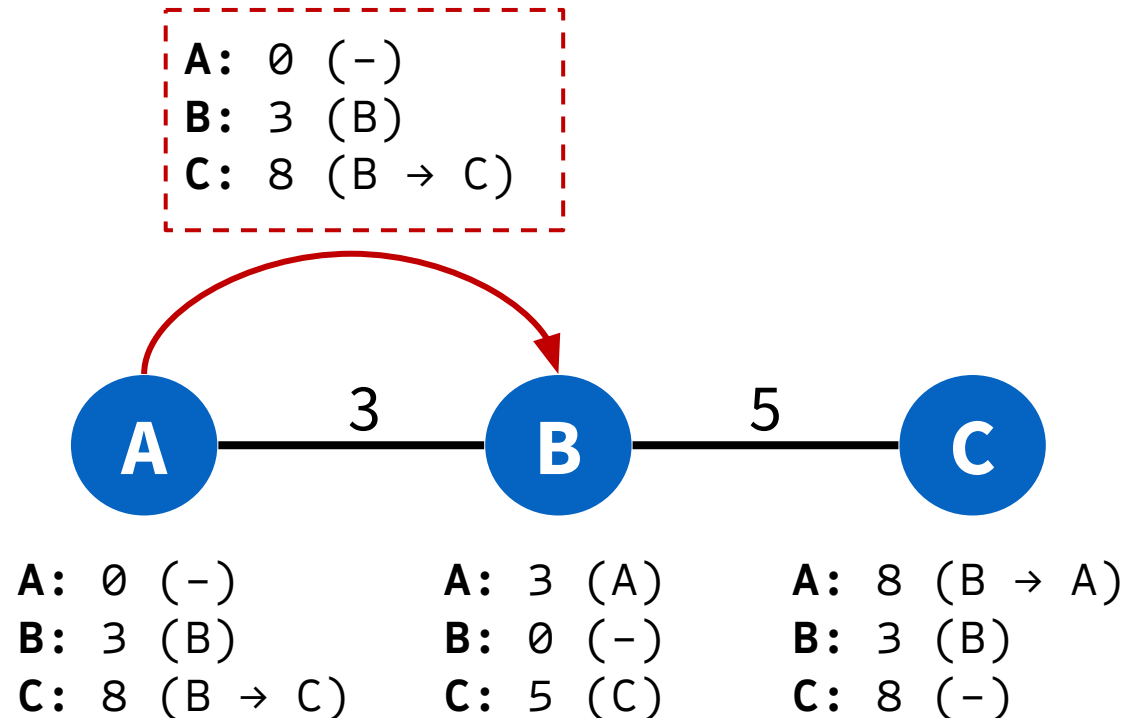
Any scenario where the loop created involves more than just a single link.



Actually Fixing Count-to-Infinity

Path-Vector protocols

- Nodes keep track of the entire path to a node along with the distance
- Messages between nodes include this entire path
- Increases size of messages/state, but routers can now prevent any routing loops



Inter-AS Routing

Q9: Inter-AS Routing

True or False: Classless addressing is more efficient than classful addressing because it has fewer options of how to divide up an IP address.

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True or False: Classless addressing is more efficient than classful addressing because it has fewer options of how to divide up an IP address.

False; classless addressing is more efficient because it allows a flexible division between the network and host part of the address. This allows a better tradeoff between the size of the routing table and effective use of the IP address space.

However, classless addressing has more options on how to divide up an IP address. This requires specifying a mask with each address as the network part of the address can be any number of bits (unlike classful addressing, which only has three options for how big the network part is).

Q10: Inter-AS Routing

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False; because BGP is focused on adhering to policies and preserving autonomy of an AS, it may not always converge to the shortest path.

Next Time

BGP Policies: How do business considerations work with routing considerations?